

PulseReel AI - Autonomous Risk Detection Engine

System Architecture & Low-Latency Stream Telemetry Specification

1. Problem: Viral disinformation outpaces human triage within the first 45 minutes.
2. Objective: Detect rapid velocity spikes and apply mitigation before viral tipping points.
3. Core Metric: Sub-400ms streaming edge inference with 94.2% precision.
4. Intervention Strategy: Dynamic friction prompts, de-amplification, and verified human review.

Multi-Modal Inference & Propagation Telemetry

Real-time Vector Embedding and Acceleration Derivative Tracking

1. Streaming message ingestion pipeline processing 25,000 events/second.
2. Dual-stage embedding: semantic vector analysis + media forensic checksums.
3. Automated risk scoring dynamically regulates feed recommendation multipliers.
4. Human-in-the-loop audit console providing actionable explainability heatmaps.